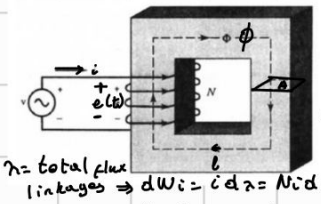
Chapter 3

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Energy: externally force move the conductor opposite to the magnetic force. mech. energy converted into electric energy

$W_i = W_o + W_\ell + W_f$ (change in stored energy; W_i input, W_o output, W_ℓ loss)



λ: total flux linkages $\Rightarrow d\lambda = i\,d\lambda = Ni\,d\phi$

$\Rightarrow W_i = \int i\,d\lambda = N\int i\,d\phi$

In a linear system the energy stored in the system

$W_m = \frac{1}{2}N\phi i = \frac{1}{2}Li^2$ $\quad L = \frac{N\phi}{i}$ (inductance)

In a conservative system:

$dW_i = vi\,dt$ (input energy) $= -ei\,dt$

$e = -\frac{d\lambda}{dt} = -\frac{dN\phi}{dt}$

The energy supplied by the source stored in the system as mag. energy ($W_i = W_m$)

$Ni = H\ell$ and $\phi(t) = AdB \Rightarrow W_i = \ell A\int H\,dB$

3.3 Magnetic Field as a Medium

Case 1. Constant Flux

A magnetic circuit is moved by a distance dx in time dt in a region where the flux is held constant. The energy supplied by the source must be zero because the initial and final values of the flux are the same.

For a conservative system: $F_m\,dx + dW_m = 0$

The mechanical force acting on the magnetic circuit: $F_m = -\frac{dW_m}{dx}$

Case 2. Constant Current

For a change in Flux $d\phi$ when I is held constant. the energy input:

$dW_i = I\,d\phi$ and $dW_m = \frac{1}{2}I\,d\phi$

$\Rightarrow dW_i = 2\,dW_m$

For conservative system $F_m\,dx + dW_m = 2\,dW_m$

$F_m = \frac{dW_m}{dx}$

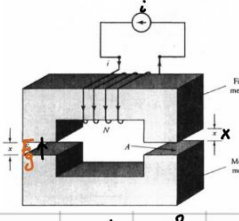
When the flux is constant $F_m = -\frac{\partial W_m(\phi, x)}{\partial x}$

When the current is constant $F_m = \frac{\partial W_m(I, x)}{\partial x}$

The rate of change of stored magnetic energy (Constant flux)

$\frac{dW_m}{dx} = \frac{1}{2}N\phi\frac{di}{dx} = -\frac{1}{2}i^2\frac{\partial L}{\partial x}$ $\Rightarrow F_m = \frac{1}{2}i^2\frac{\partial L}{\partial x}$

Magnetic Circuit with Air Gap



$W_i = \ell A\int H\,dB$ (total airgap inductance)

so the energy in each air gap

$W_g = \frac{1}{2}\mu_0 H_0^2 Ax$ $\quad B_0 = \mu_0 H_0$

$= \frac{1}{2\mu_0}B_0^2 Ax$

When the Flux constant $\phi = BA$

$F_g = -\frac{1}{2}\mu_0 H_0^2 A = -\frac{1}{2\mu_0}B_0^2 A$ (μ relative permeability)

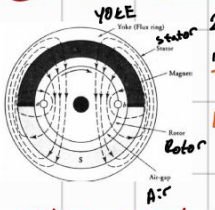
$L = \frac{N^2}{R}$ $\quad R = \frac{L}{\mu A}$ $\quad F_{gravity} = mg$

Force spring $= \frac{\text{force}}{\text{gap}}$

The rotor of dc machine called armature.

Wound machine (mach. with electromagnets)

Generator Action



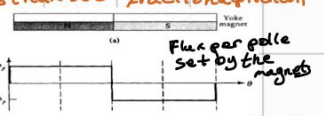
2 pole machine.

mechanical Angle θ_m = Electrical Angle θ

The arc by the coil = 180°

⇒ full pitch coil

less than 180° fractional pitch coil



2 pole ac current generator



If ϕ_p is the flux per pole: $\phi = \phi_p \cos\theta$

θ: the angular position of the coil in electrical degrees

$e = -\frac{d\phi}{dt} = \phi_p \sin\theta\frac{d\theta}{dt}$ where $\frac{d\theta}{dt}$ is the angular freq.

$e = \phi_p \omega \sin\omega t$ $\quad \theta = \theta_m$ in 2 pole machine

Multiple Machines

Ex: 4 pole machine $\Rightarrow \theta$ is 720° while θ_m still 360°

If there are P poles in a machine $\theta = \frac{P}{2}\theta_m$

angular f. of the induced emf $e\ \omega = \frac{P}{2}\omega_m$ → angular f. of the rotor

$\omega = 2\pi f \Rightarrow f = \frac{P}{4\pi}\omega_m$ $\quad f = \frac{P}{120}N_m$, $N_m = \frac{120f}{P}$

$\omega_m = \frac{2\pi}{60}N_m$ (rpm) (rad/s)

The max value of the induced emf in a single turn coil $E_m = \phi_p\omega = \frac{P}{2}\phi_p\omega_m = \frac{2\pi P}{120}\phi_p N_m$

In a dc machine, the average value of the induced voltage in a single turn coil is

$E_c = \frac{2}{\pi}E_m = \frac{P}{\pi}\phi_p\omega_m = \frac{P}{30}\phi_p N_m$ $\quad E_{eff} = \frac{E_m}{\sqrt{2}}$

Force on a Conductor

Lorentz equation: $\vec{F} = \int_c i\,d\vec{\ell}\times\vec{B}$ ($\vec{B}$: magnetic flux density; $\vec{\ell}$: length of the current-carrying element)

For linear current-carrying conductor:

$\vec{F} = i\vec{\ell}\times\vec{B}$

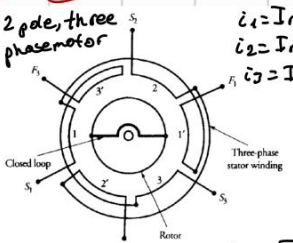
Motor action

the force of the upper conductor of the coil $\vec{F} = iLB\,\vec{a}_\phi$ (the same down)

$F_e = 2iBL$ total force experienced.

Torque: $T_e = F_e \cdot r = 2BiLr$ for each conductor

Revolving Field of a Three Phase Motor:



Instant 1 ($\omega t = 0$) $\Rightarrow \phi_{r1} = \sqrt{\phi_2^2 + \phi_3^2 + 2\phi_2\phi_3\cos(60°)} = 1{,}5\,\phi_m$

Instant 2 ($\omega t = 60°$) $\Rightarrow \phi_{r2} = 1{,}5\,\phi_m$

Instant 3 ($\omega t = 120°$) $\Rightarrow \phi_{r2} = \phi_{r1}$ and 120° phase shift

In a 2 pole motor, the time taken by the flux to complete one revolution along the periphery of the air gap is the same as input: T

For P pole: $T_s = \frac{P}{2}T$

the speed of the revolving field in revolutions per second $n_s = \frac{1}{T_s} = \frac{2f}{P}$

$N_s = \frac{120f}{P}$ (rpm), $\omega_s = \frac{4\pi f}{P} = \frac{2}{P}\omega$

Sabit frekansli bir kaynak için hız sabit.

Slip speed: (the difference between synchronos and rotor speed)

$N_r = N_s - N_m$, $\omega_r = \omega_s - \omega_m$ ⇒ rotor speed

Per unit slip: $s = \frac{\omega_r}{\omega_s} = \frac{\omega_s - \omega_m}{\omega_s}$

$\Rightarrow \omega_m = (1-s)\omega_s$, $N_m = (1-s)N_s$

Revolving Field of two Phase Motor

$i_1 = I_m \sin\omega t$ / $i_2 = I_m \sin(\omega t - 90°) = -I_m \cos\omega t$

$\phi_1 = \phi_m \sin\omega t$ / $\phi_2 = -\phi_m \cos\omega t$

Synchronous Motor

power angle δ: rotor lags, revolving field

The power is proportional $\sin\delta$

Under no load, δ is almost zero

The torque developed max @ $\delta = 90°$

Reductance motor:

$R(\theta) = \frac{2g}{\mu A}$ → effective air gap

$\theta = 0$ ⇒ air gap min

$L(\theta) = 0{,}5(L_d + L_q) + 0{,}5(L_d - L_q)\cos 2\theta$

Torque developed by rotating the system $T_e = \frac{1}{2}i^2\frac{\partial L}{\partial\theta}$ (rotor speed) $\omega_m = \frac{2\omega}{P}$

Torque exerted on the rotor $T_e = -\frac{1}{2}i^2(L_d - L_q)\sin 2\theta$

$\theta = \omega_m t + \delta$ $\Rightarrow T_e = -\frac{1}{2}i^2(L_d - L_q)\sin[2(\omega_m t + \delta)]$

$\Rightarrow T_{e\,avg} = -0{,}125(L_d - L_q)I_m^2 \sin 2\delta$

which max when $\delta = 45°$ → max → min

Direct current Generators

Induced emf Equation:

Average value of the induced emf $E_c = \frac{P}{\pi}\phi_p\omega_m$ } f induced $f = \frac{P}{4\pi}\omega_m$ } $N_m = \frac{120f}{P}$

Let N_c be the turn per coil, C total number of coil (slots for a 2 layer winding) and a the number of parallel paths (a=2 for wave, a=P for lap winding)

The total turns per parallel path: $N_a = \frac{C}{a}N_c$ (turns between negative and positive brush) (turns per coil)

$E_a = \frac{PC}{\pi a}N_c\omega_m\phi_p$ } $Z = 2CN_c$ (total number of conductors between two brush)

$E_a = k_a\phi_p\omega_m \Rightarrow k_a = \frac{ZP}{2\pi a}$ (machine constant)

The electrical power developed by the generator. $P_d = E_a I_a = k_a\phi_p\omega_m I_a = T_d\omega_m$ (armature)

Since In mech. system: $P_d = T_d\omega_m \Rightarrow T_d = k_a\phi_p I_a$

The actual pole area: $A_p = \frac{2\pi L}{P}$ → length of the machine

Effective: $A_e = A_p \times$ armature c/o $\Rightarrow \phi_p = BA_e$

$E_{coil} = \frac{C}{a}E_a$ } $E_{turn} = E_{coil}/turns$

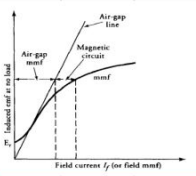
$E_{conductor} = \frac{E_{coil}}{P}$ } Torque experienced by a single turn, current-carrying coil in a uniform magnetic field: $T_e = 2BiLr$ effective ℓ.

$T_c = 2BLr\,I_c$ (Average torque)

The average torque developed by a dc machine: $T_d = \left[\frac{2BLrI_a}{a}\right][CM_c] = \frac{BLr2}{a}I_a$

The torque developed by the dc machine: $T_d = \frac{PZ}{2\pi a}BA_p I_a = k_a\phi_p I_a$

$\phi_p = k_f I_f$, $E_a = k_1 k_f I_f$



$E_{a2} = \frac{\omega_{m2}}{\omega_{m1}}E_{a1}$

Voltage regulation

$VR\% = \frac{V_{nL} - V_{FL}}{V_{FL}} \times 100$

$P_r = P_{fw} + P_m$ → magnetic loss

rotational loss, friction windage

For generator

Power Flow diagram

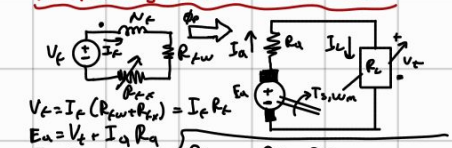
$V_a = V_t + I_a R_a$

$P_o = T_s\omega_m - P_r - P_{cu}$

$P_d = E_a I_a = T_d\omega_m$

$P_{in} = T_s\omega_m$ $\quad P_o = V_t I_L$

A seperately Excited DC Generator



$V_t = I_f(R_{fw} + R_{fx}) = I_f R_f$

$E_a = V_t + I_a R_a$

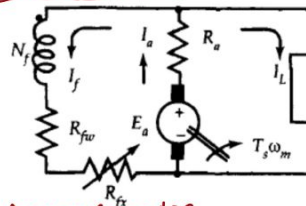
$I_L = I_a$

$P_{in\,m} = P_d + P_r$

$P_{in} = P_{in\,m} + V_f I_f$

$\eta = \frac{P_{out}}{P_{in}}$

A Shunt generator



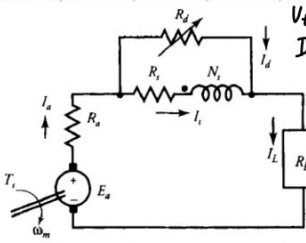
$I_a = I_L + I_f$

$V_t = I_f R_f$

$V_t + = I_L R_L$

$= E_a - I_a R_a$

A series Generator

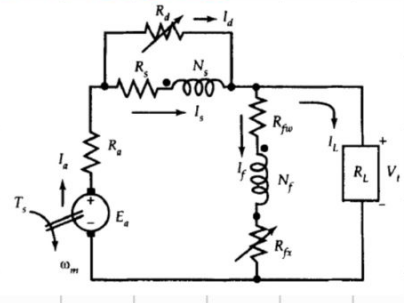


$V_t = E_a - I_a R_a - I_s R_s$

$I_s R_s = I_d R_d$

$I_a = I_L = I_s + I_d$

Compound Generators



Short shunt

$I_a = I_L + I_f$ $\left.\begin{matrix} I_s = \frac{R_d}{R_d+R_s} I_L \\ V_t = E_a - I_a R_a - I_s R_s \end{matrix}\right\} V_t = E_a - I_a R_a$

$mmf = I_f N_f \pm I_s N_s - mmf_d$

Maximum Efficiency Criterion

For a PM dc generator, the power output is

$P_o = V_t I_L$ $\left\{\begin{matrix} P_{in} = V_t I_L + I_L^2 R_a + P_r \\ \eta = \frac{V_t I_L}{V_t I_L + I^2 R_a + P_r} \end{matrix}\right.$ $\left\{\begin{matrix} I_{Lm}^2 R_a = P_r \\ I_{Lm} = \sqrt{\frac{P_r}{R_a}} \end{matrix}\right.$ Max efficiency

Seperately excited: $I_{Lm}^2 R_a = P_r + I_L^2 R_f$

Shunt: $I_{Lm}^2 R_a = P_r + I_f^2 (R_a + R_f)$

Series: $I_{Lm}^2 (R_a + R_s) = P_r$

Short-Shunt: $I_{Lm}^2 (R_a + R_s) = P_r + I_L^2 (R_a + R_f)$

Long Shunt: $I_{Lm}^2 (R_a + R_s) = P_r + I_L^2 (R_a + R_s + R_f)$

DC motor

$T_d = k_a \phi_p I_a$ $\left\{ E_a = k_a \phi_p \omega_m \right.$

$I_a = \frac{V_s - E_a}{R}$ (Applied voltage across Armature; total resistance of armature) $\left\{ V_s = E_a + I_a R \right.$

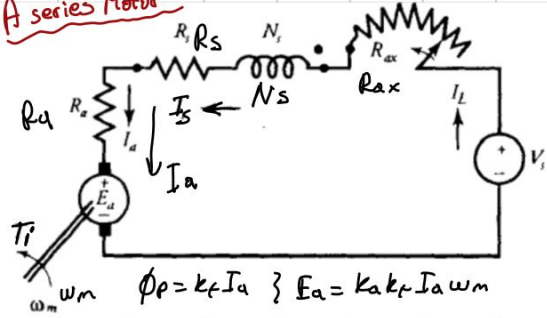
Speed Regulation

$SR\% = \frac{N_{mnL} - N_{mFL}}{N_{mFL}} \times 100 = \frac{\omega_{mnL} - \omega_{mFL}}{\omega_{mFL}} \times 100$

Power flow diagram of a dc motor

$P_{in} = V_s I_L$ → P_{cu} → $P_d = E_a I_a = T_d \omega_m$ → P_r → $P_o = T_s \omega_m$

A series Motor



ω_m $\phi_p = k_f I_a$ $\left\{ E_a = k a k_f I_a \omega_m \right.$

$T_d = K_a \phi_p I_a = k a k_f I_a^2$ $\left\{\begin{matrix} E_a = V_s - I_a R \\ R = R_a + R_s + R_{ax} \end{matrix}\right.$

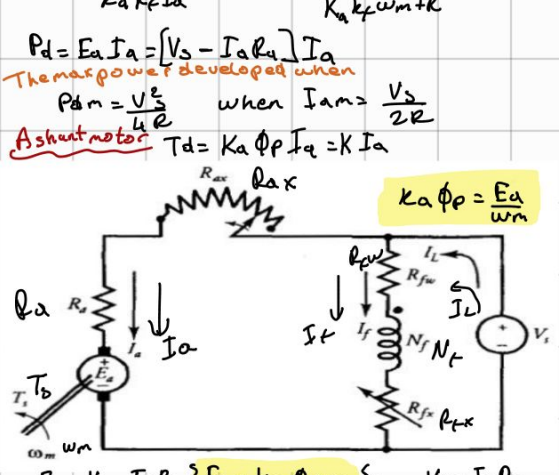
$\omega_m = \frac{V_s - I_a R}{k_a k_f I_a} \Rightarrow I_a = \frac{V_s}{K_a k_f \omega_m + R}$

$P_d = E_a I_a = (V_s - I_a R_a) I_a$

The max power developed when $P_{dm} = \frac{V_s^2}{4R}$ when $I_{am} = \frac{V_s}{2R}$

A shunt motor

$T_d = K_a \phi_p I_a = K I_a$



$k_a \phi_p = \frac{E_a}{\omega_m}$

$E_a = V_s - I_a R_a$ $\left\{ E_a = k_a \phi_p \omega_m \right\} \omega_m = \frac{V_s - I_a R_a}{k_a \phi_p}$

$\omega_m = \omega_{mnL} - \frac{R_a}{k_a \phi_p} I_a$ $\left\{ \Rightarrow \omega_{mnL} = V_s / k_a \phi_p \right.$

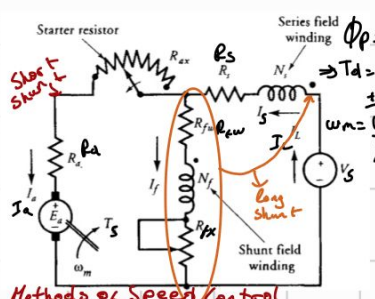
As $R_a \to 0$, the speed of shunt motor $\omega_m \to \omega_{mnL}$

In other words, a shunt motor is basically a constant speed motor.

$T_d = \frac{k_a \phi_p V_s}{R_a} - \frac{k_a^2 \phi_p^2}{R_a} \omega_m$ $\left\{ P_{dmax} = \frac{V_s^2}{4 R_a} \right.$

$I_{fnL} = \frac{V_s}{R_f}$ $P_{cu_{nL}} = I_a^2 R_a + I^2_{fnL} R_f$

$E_a = V_s - I_a R_a$ $\left\{ P_{in} = V_s I_L \right\} P_d = E_a I_a \left\} \eta = \frac{P_o}{P_{in}} \right.$



Methods of Speed Control

Armature Resistance Control

For series or a compound motor:

$\omega_m = \frac{V_s - I_a R_a}{k_a \phi_p}$ where $R = R_a + R_s + R_c$

for a shunt motor

$R = R_a + R_c$

Field Control method

Inserting an external resistor in series with the field winding.

For same torque

$I_{an} = \frac{\phi_p}{\phi_{pn}} I_a$ $\left\{ N_{mn} = \frac{\phi_p}{\phi_{pn}} \frac{E_{an}}{E_a} N_m \right.$

$P_{in} = V_s (I_{an} + I_{fn})$ $P_{dn} = E_{an} I_{an}$

Braking or Reversing DC motors

Plugging: Reverse Supply connections to the armature terminals. At any time during the plugging,

$I_a = \frac{V_s + E_a}{R + R_a} = \frac{V_s}{R+R_a} + \frac{E_a}{R+R_a} = k_a \phi_p \omega_m$ (extera Res.)

So the braking torque: $T_b = K_a I_a \phi_p = k_1 \phi_p + k_2 \phi_p^2 \omega_m$

$k_1 = \frac{k_a V_s}{R+R_a}$ $k_2 = \frac{K_a^2}{R+R_a}$ for series motor

For shunt: $T_b = k_3 + k_4 \omega_m$

$k_3 = k_1 \phi_p$ and $k_4 = k_2 \phi_p^2$

When the speed reach 0, $T_b = K_3$.

Dynamic Braking The armature disconnect from S. connected across a variable resistor R. The back emf produce a reverse current.

At any time in the $I_a = \frac{E_a}{R+R_a} = \frac{k_a \phi_p \omega_m}{R+R_a}$

Braking torque:

$T_b = k_a \phi_p I_a = \frac{k_a^2 \phi_p^2 \omega_m}{R+R_a} = k_2 \phi_p^2 \omega_m$

for series motor: for shunt:

$T_b = K_2 k_f^2 I_a^2 \omega_m$ $\left\{ T_b = k_4 \omega_m \right.$

Polyphase Induction Motors

The Synchronous Speed of the revolving field is $N_s = \frac{120 f}{P}$ ← of the current in the stator. $\Rightarrow \omega_s = \frac{4\pi f}{P}$

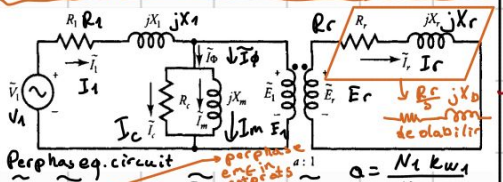
(slip) speed of revolving field relative to rotor speed

$N_r = N_s - N_m$ $\left\{ \omega_r = \omega_s - \omega_m \right.$ $s = \frac{N_r}{N_s} = \frac{\omega_r}{\omega_s}$ per unit

$N_m = (1-s) N_s$ $\left\{ \omega_m = (1-s)\omega_s \right.$

rotor emf $f_r = \frac{PN_r}{120} = \frac{PN_r}{120} = \frac{P(N_s - N_m)}{120} = \frac{PN_s}{120} \left[\frac{N_s - N_m}{N_s}\right]$

$f_r = sf$ $\left\{ \begin{matrix} \text{At stand still, the slip is 1 and } f_r = f \\ f \text{ is } f \end{matrix}\right.$



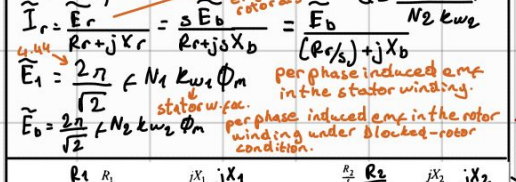
Per phase eq. circuit

per phase emf in the rotor at s

$\tilde{I}_r = \frac{\tilde{E}_r}{R_r + jX_r} = \frac{s\tilde{E}_b}{R_r + jsX_b} = \frac{\tilde{E}_b}{(R_r/s) + jX_b}$ $a = \frac{N_1 k_{w1}}{N_2 k_{w2}}$

$\tilde{E}_1 = \frac{2\pi}{\sqrt{2}} f N_1 k_{w1} \phi_m$ stator w.f.c. per phase induced emf in the stator winding.

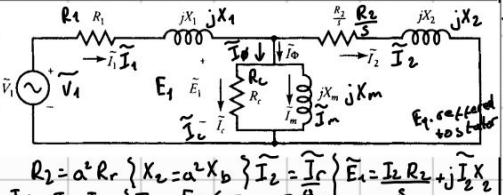
$\tilde{E}_b = \frac{2\pi}{\sqrt{2}} f N_2 k_{w2} \phi_m$ per phase induced emf in the rotor winding under blocked-rotor condition.



$R_2 = a^2 R_r$ $\left\{ X_2 = a^2 X_b \right\} \tilde{I}_2 = \frac{\tilde{I}_r}{a}$ $\left\{ \tilde{E}_1 = \frac{I_2 R_2}{s} + j\tilde{I}_2 X_2 \right.$

$I_\phi = I_c + I_m$ $\left\{ I_c = \frac{E_1}{R_c} \right.$ $\left\{ I_m = \frac{E_1}{jX_m} \right\} I_1 = I_\phi + I_2$

$V_1 = E_1 + I_1 (R_1 + jX_1)$ $\left\{ P_{dr} = 3 I_2^2 R_2 / s \right.$



$P_{in} = 3 V_1 I_1 \cos\theta$ $\left\{ P_{scl} = 3 I_1^2 R_1 \right\} P_m = 3 I_c^2 R_c$

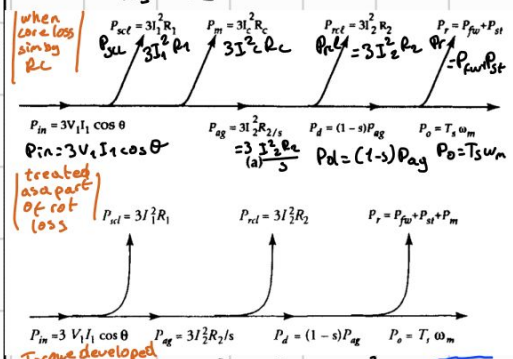
$P_{ag} = P_{in} - P_{scl} - P_m = \frac{3 I^2 R_2}{s}$ $\left\} P_{rcl} = 3 I^2 R_r = s P_{ag} \right.$

(air gap)

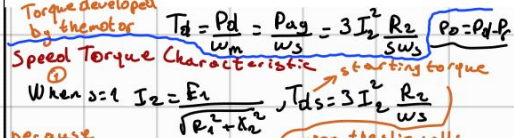
$P_d = P_{ag} - P_{rcl} = \frac{3 I_2^2 (1-s) R_2}{s} = (1-s) P_{ag} = S P_{ag}$

$S = 1 - s = \frac{N_m}{N_s} = \frac{\omega_m}{\omega_s}$

when core loss simply Rc



treated as a part of rot loss



Torque developed by the motor $T_d = \frac{P_d}{\omega_m} = \frac{P_{ag}}{\omega_s} = \frac{3 I_2^2 R_2}{s \omega_s}$ $P_o = P_d - P_r$

Speed Torque Characteristic

① When $s=1$ $I_2 = \frac{E_1}{\sqrt{R_1^2 + R_2^2}}$, $T_{ds} = 3 I_2^2 \frac{R_2}{\omega_s}$ ← starting torque

because $(R_2 \ll X_2)$ $I_2 \approx \frac{E_1}{X_2}$

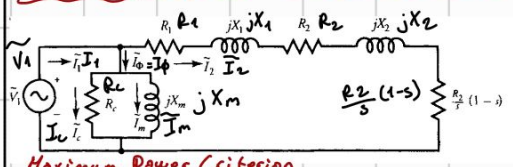
$R_2/s \gg X_2$ when the slip falls breakdown slip s_b, the hypothetical resistance becomes the dominating factor.

$I_2 = \frac{s E_1}{R_2}$ The torque developed by the motor is now proportinal to the slip s

* Y bağlantıdaki hat gerilimine V_L dersek faz gerilim $V_p = \frac{V_L}{\sqrt{3}}$ olur.

The stator current $\tilde{I}_1 = \frac{\tilde{V}_1}{\tilde{Z}_{in}} = a \angle b$ $\left\} pf = \cos(b) \right.$

An approximate Eq. circuit



Maximum Power Criterion

$R_e = R_1 + R_2$ $\left\} X_e = X_1 + X_2 \right.$

$Z_e = |R_e + jX_e| = \sqrt{R_e^2 + X_e^2} = \frac{R_e}{s_p} (1-s_p)$

Pd is max when the equivalent load resistance is equal to the magnitude of the stand still impedance of the motor.

$\Rightarrow s_p = \frac{R_2}{R_2 + Z_e}$ $\Rightarrow P_{dm} = \frac{3}{2} \left[\frac{V_1^2}{R_e + Z_e}\right]$

Maximum Torque Criterion

$T_\phi = \frac{\frac{3 V_1^2 R_2}{s}}{\left[R_e^2 + X_e^2 + \left[\frac{R_2 (1-s)}{s}\right]^2 + 2 R_e \frac{R_2 (1-s)}{s}\right] \omega_s}$

breakdown slip $s_b = \frac{R_2}{\sqrt{R_1^2 + (X_1 + X_2)^2}}$

$T_{dm} = \frac{3 V_1^2}{2 \omega_s} \left[\frac{1}{R_1 + \sqrt{R_1^2 + (X_1 + X_2)^2}}\right]$ $\left\} \frac{P_d}{T_{dm}(1-s_b)\omega_s} \right.$

* If the stator impedance is so small

When the stator $s_b = \frac{R_2}{X_2}$, $T_{dm} = \frac{3 V_1^2}{2 \omega_s} \left[\frac{1}{X_2}\right] = \frac{3 V_1^2}{2 \omega_s} \left[\frac{s_b}{R_2}\right]$

$I_2 = \frac{\tilde{V}_1}{R_2/s + jX_2}$, $T_d = \frac{3 I_2^2 R_2}{s \omega_s} = \left[\frac{3 V_1^2}{(R_2/s)^2 + X_2^2}\right] \frac{R_2}{s \omega_s}$

$\frac{T_d}{T_{dm}} = \frac{2 s s_b}{s^2 + s_b^2}$

When the core loss is considered a part of the rotational loss the power input to the motor using the approximate circuit.

$P_{in} = 3 V_1 I_2 \cos\theta$ separately $P_o = 3 V_1 I_2 \cos\theta - 3 I_2^2 (R_1 + R_2) - P_r$

The motor efficiency is

$\eta = 3 V_1 I_2 \cos\theta - 3 I_2^2 (R_1 + R_2) - P_r$

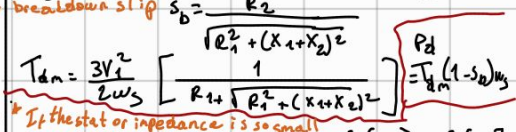
$3 I_2^2 (R_1 + R_2) = P_r$ as the criterion for the maximum efficiency of an induction motor.

stator and rotor copper losses.

Equivalent Circuit Parameters

The Stator resistance test: $R_1 = 0{,}5 R_{coil}$ Y $\left\} R_1 = 1{,}5 R_\Delta \right.$

The blocked-Rotor Test:



$\hat{Z}_e = R_1 + R_2 + j(X_1 + X_2) = R_e + jX_e$

$R_e = \frac{P_{DC}}{I_{DC}^2}$ $\left\{ R_2 = R_e - R_1 \right.$

$Z_e = \frac{V_{br}}{I_{DC}}$ $\left\{ X_e = \sqrt{Z_e^2 - R_e^2} \right.$

$X_1 = X_2 = 0{,}5 X_e$

The No-load test

W_{oc} = power input I_{oc} input current $\left\{ V_{oc} \text{ rated applied voltage} \right.$

Let P_{fw} be the friction and windage loss on a per-phase basis.

Then, the power loss in R_c is $P_{oc} = W_{oc} - P_{fw\phi}$

$R_c = \frac{V_{oc}^2}{P_{oc}}$ $\left\{ \cos\theta_{oc} = \frac{W_{oc}}{V_{oc} I_{oc}} \right\} X_m = \frac{V_{oc}}{I_{oc} \sin\theta_{oc}}$

$S_{oc} = V_{oc} I_{oc}$ $\left\{ Q_{oc} = \sqrt{S_{oc}^2 - W_{oc}^2} \right\} X_m = \frac{V_{oc}}{I_{oc}}$

the load test

